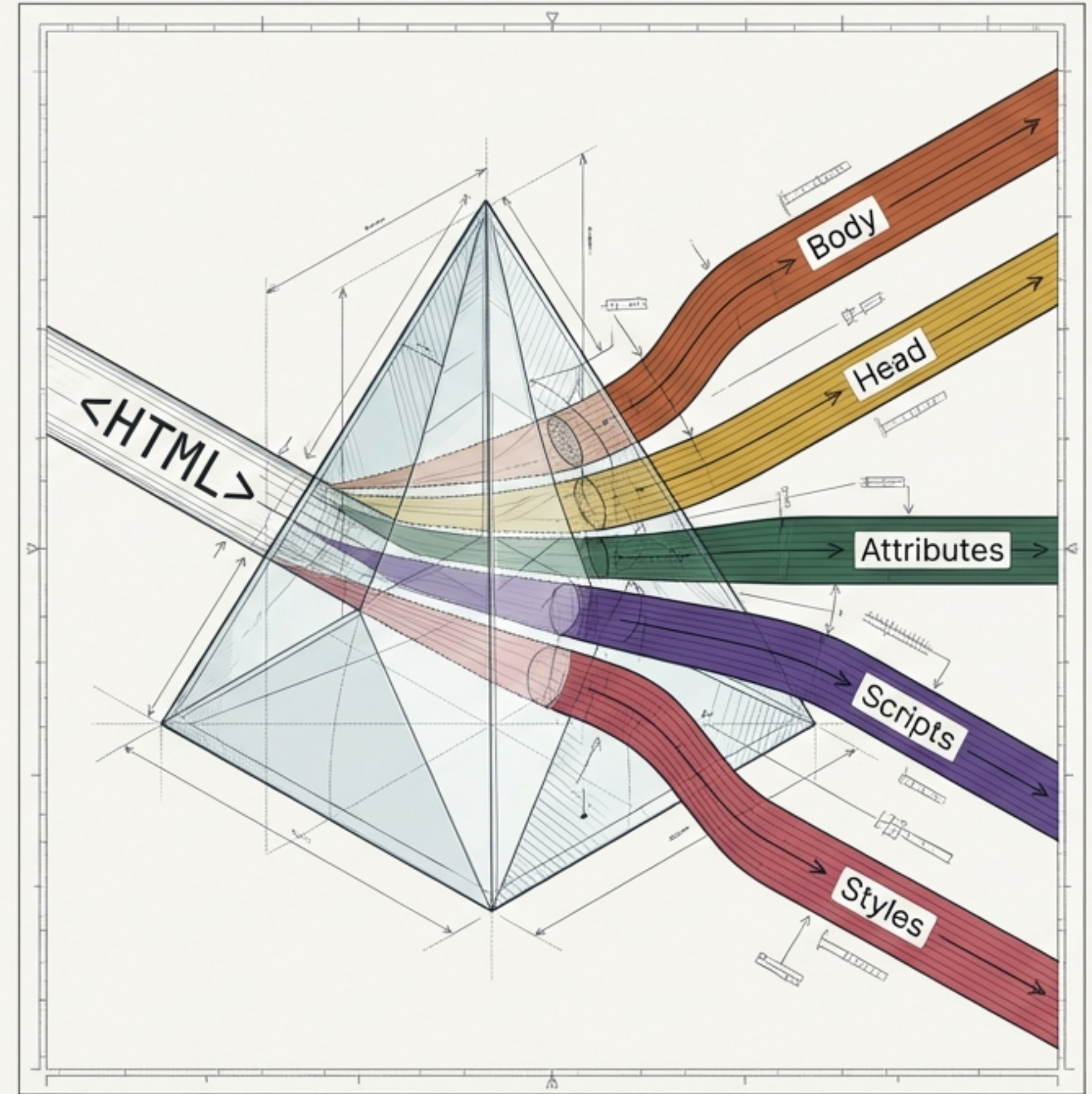


Html_MGraph

Database Normalisation for the Web

Transforming HTML documents into interconnected, type-safe intelligence. A multi-graph architecture that separates structure, style, and logic while maintaining strict relationships.



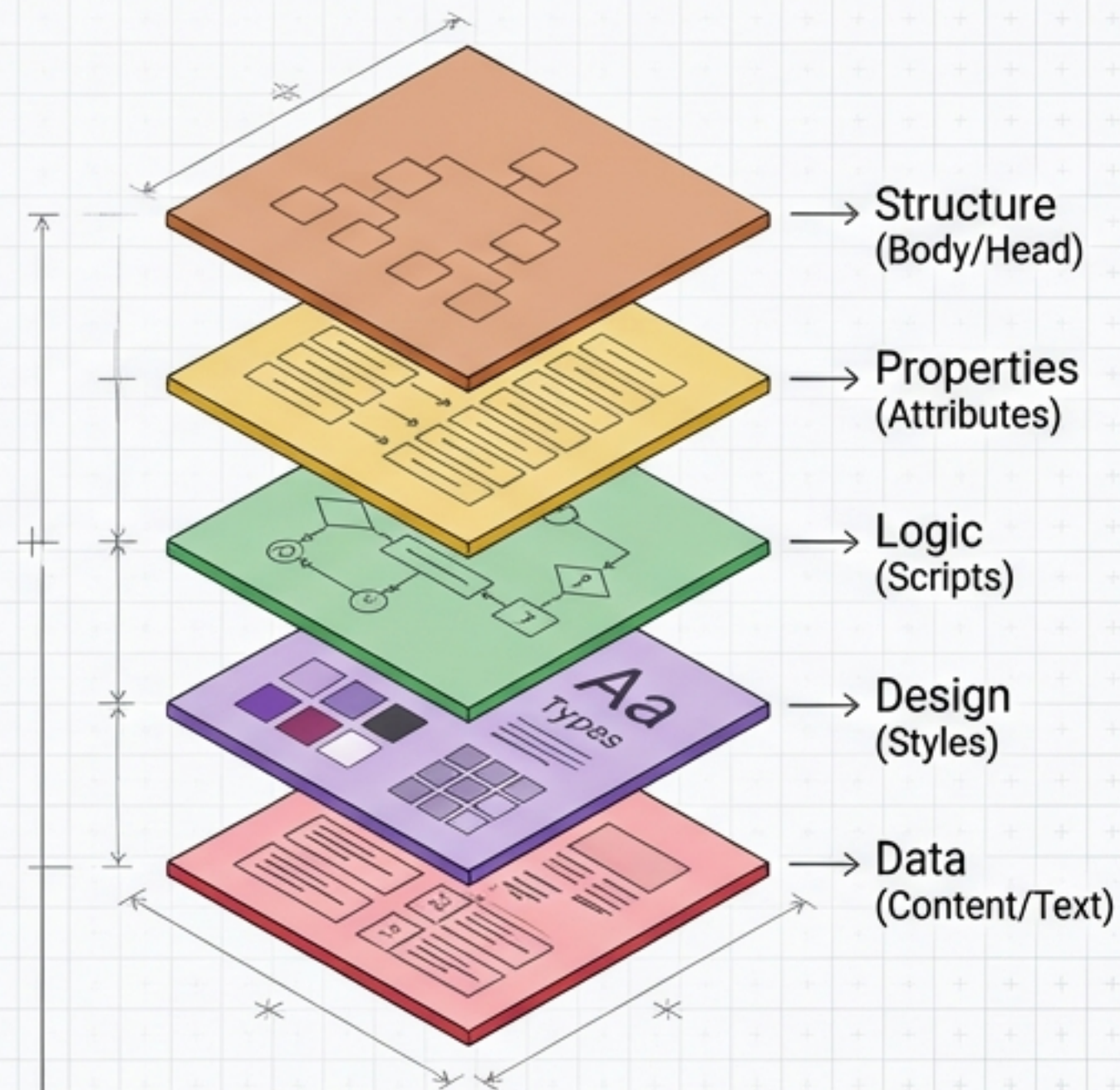
MOVING BEYOND THE CLUTTERED DOM

THE MONOLITH TRADITIONAL DOM PARSING



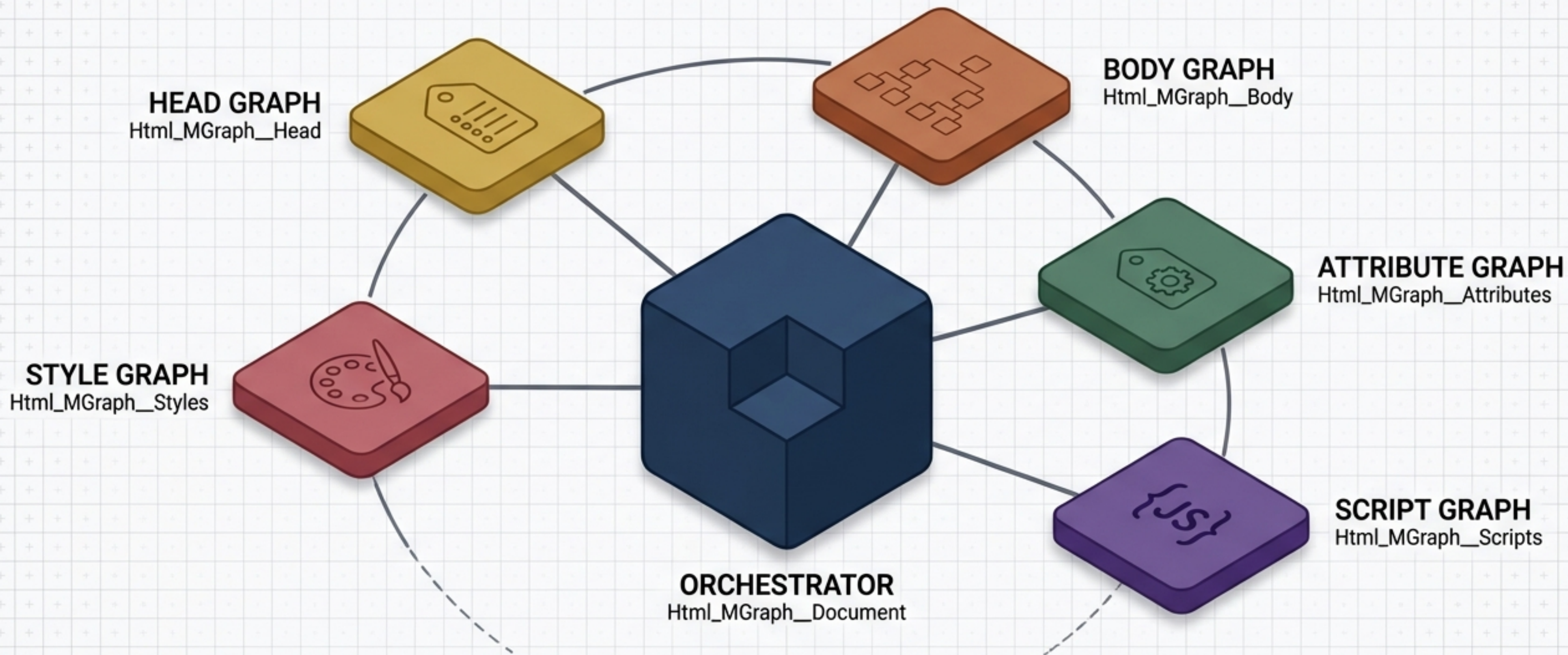
Mixed Concerns & Attribute Noise

THE DECONSTRUCTED VIEW HTML_MGRAPH ARCHITECTURE



Separation of Concerns & Type-Safety

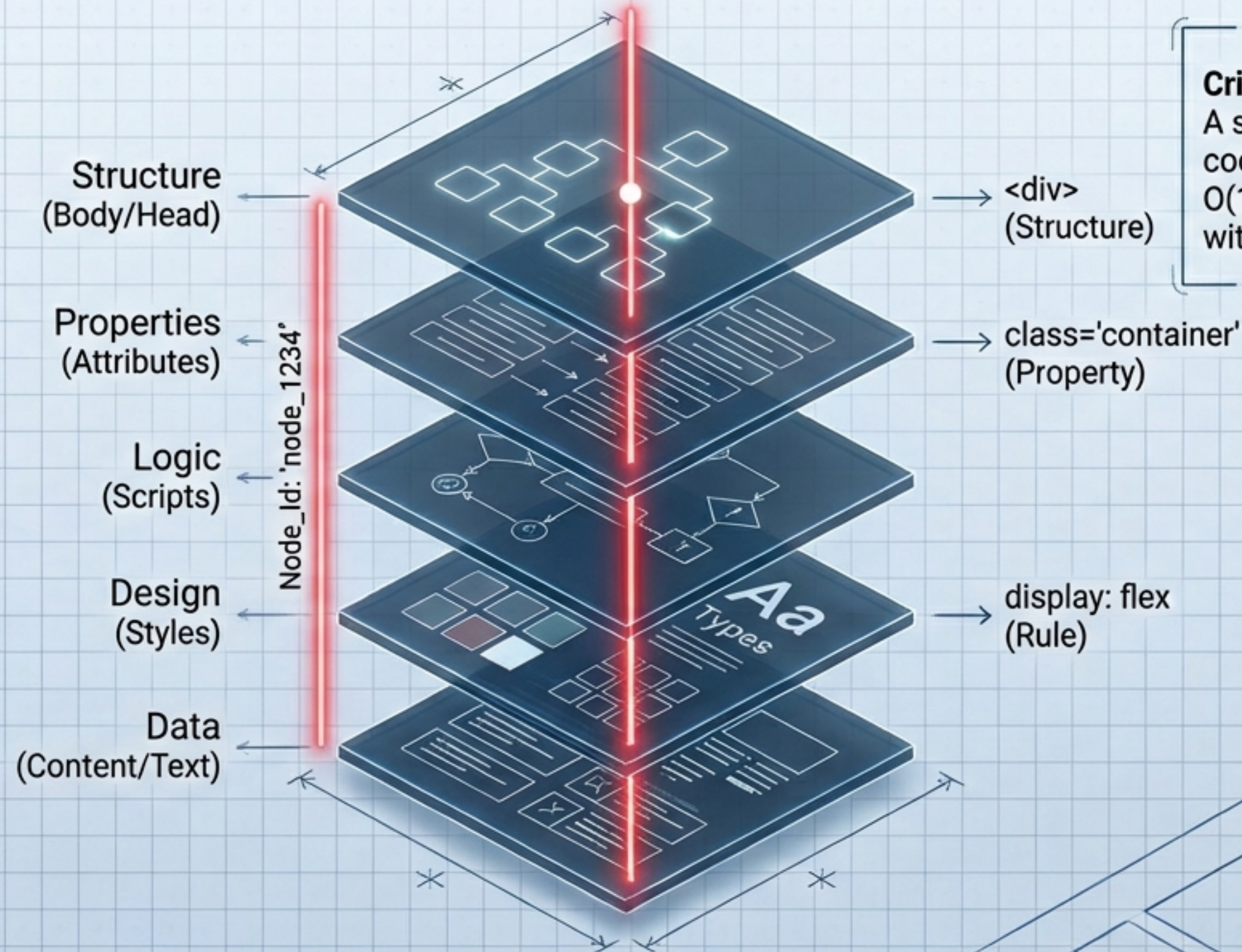
THE 5+1 GRAPH ECOSYSTEM



The Orchestrator holds the reference to the ecosystem, allowing specialized workers to manage distinct aspects of the HTML document.

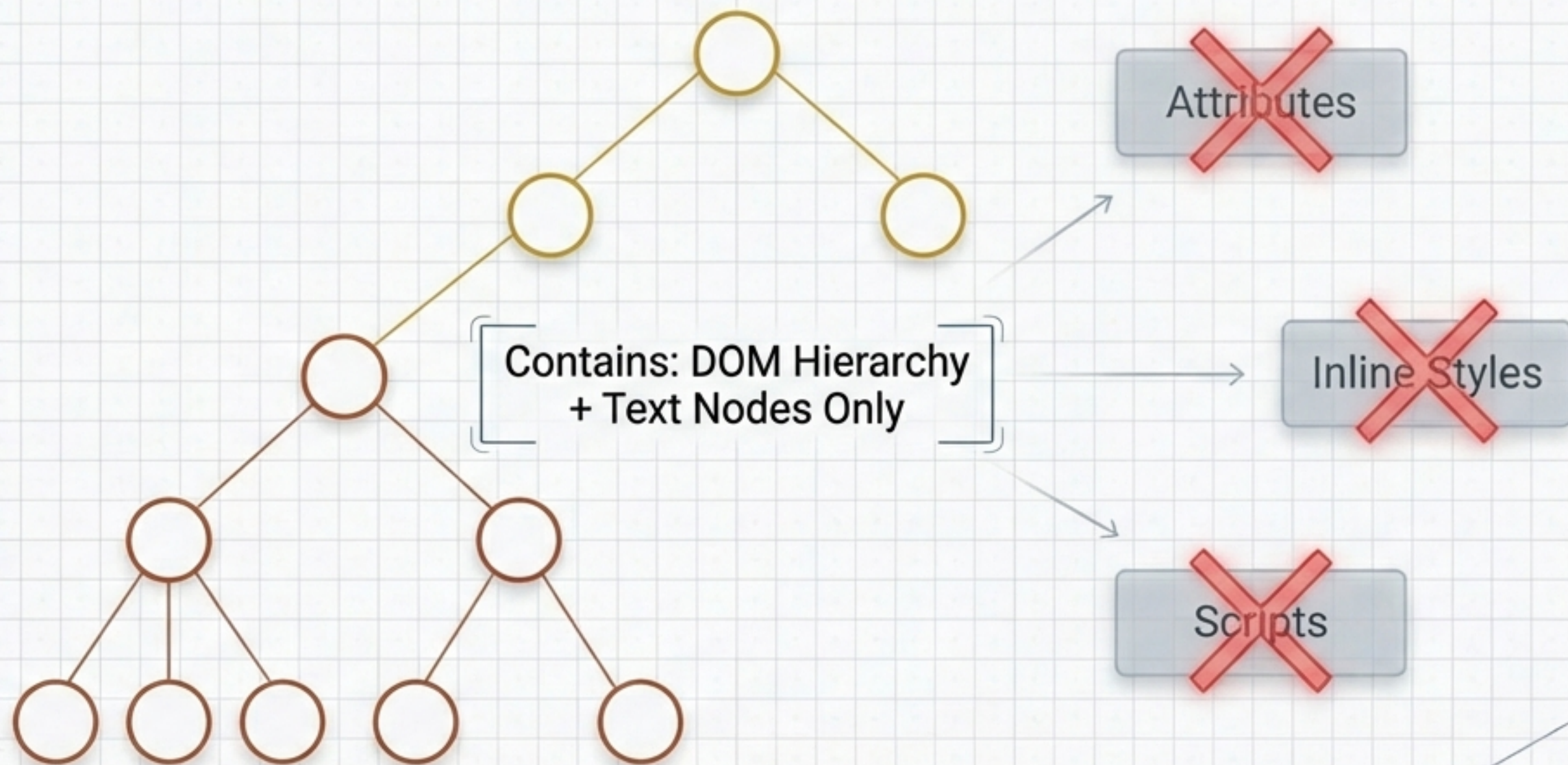
UNIFIED IDENTITY ACROSS DIMENSIONS

The Shared Node_Id Pattern



Pure Structure, Zero Noise

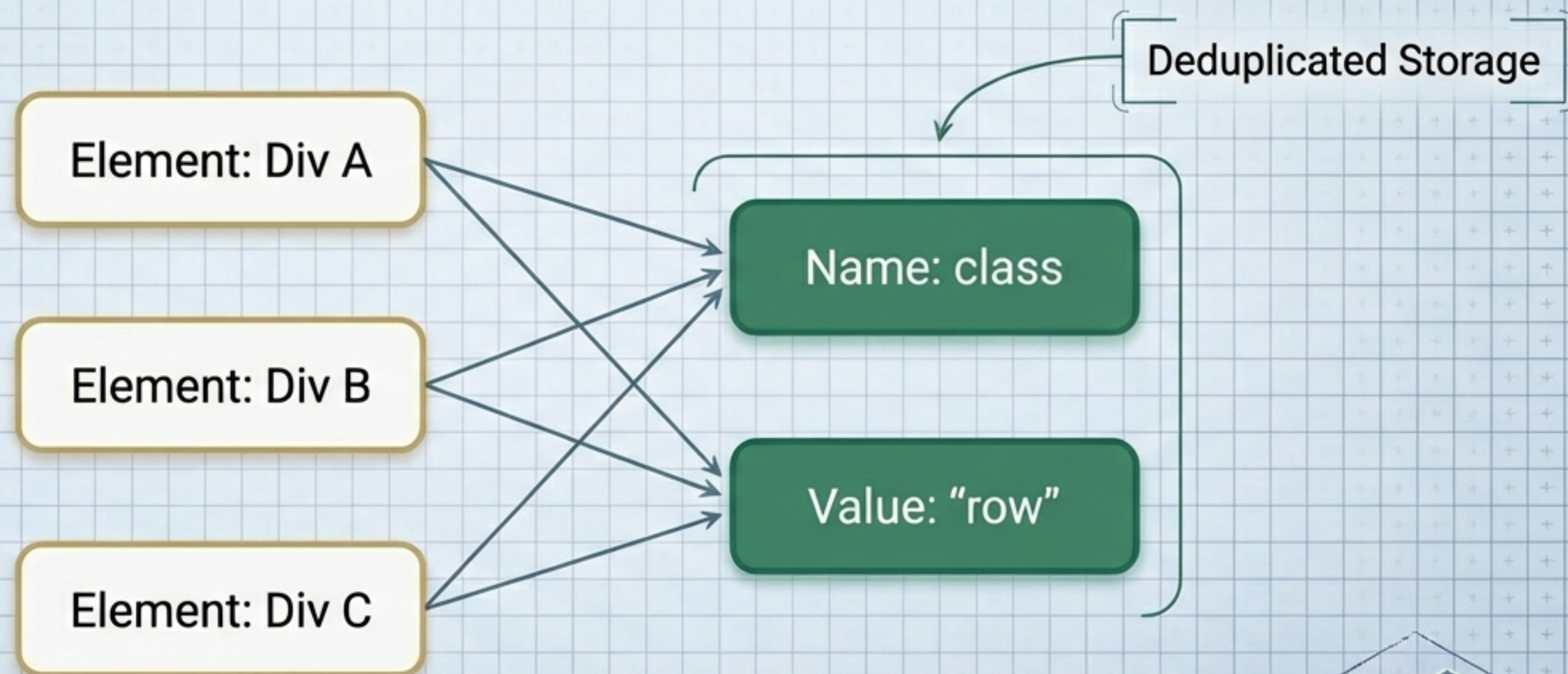
Html_MGraph_Head & Html_MGraph_Body



Query the tree shape without implementation details.
Traversal is optimized for structural analysis.

The Optimized Attribute Engine

Html_MGraph_Attributes



The Three-Node Model separates Tag, Name, and Value. Caching enables $O(1)$ tag lookups and significant memory efficiency for repetitive HTML structures.

Isolating Behavior and Aesthetics

Html_MGraph_Scripts & Html_MGraph_Styles

Scripts Graph

```
...  
Node_Id: "node_55"  
Type: "text/javascript"  
Source: Inline  
Content: "console.log('Logic isolated');"  
...
```

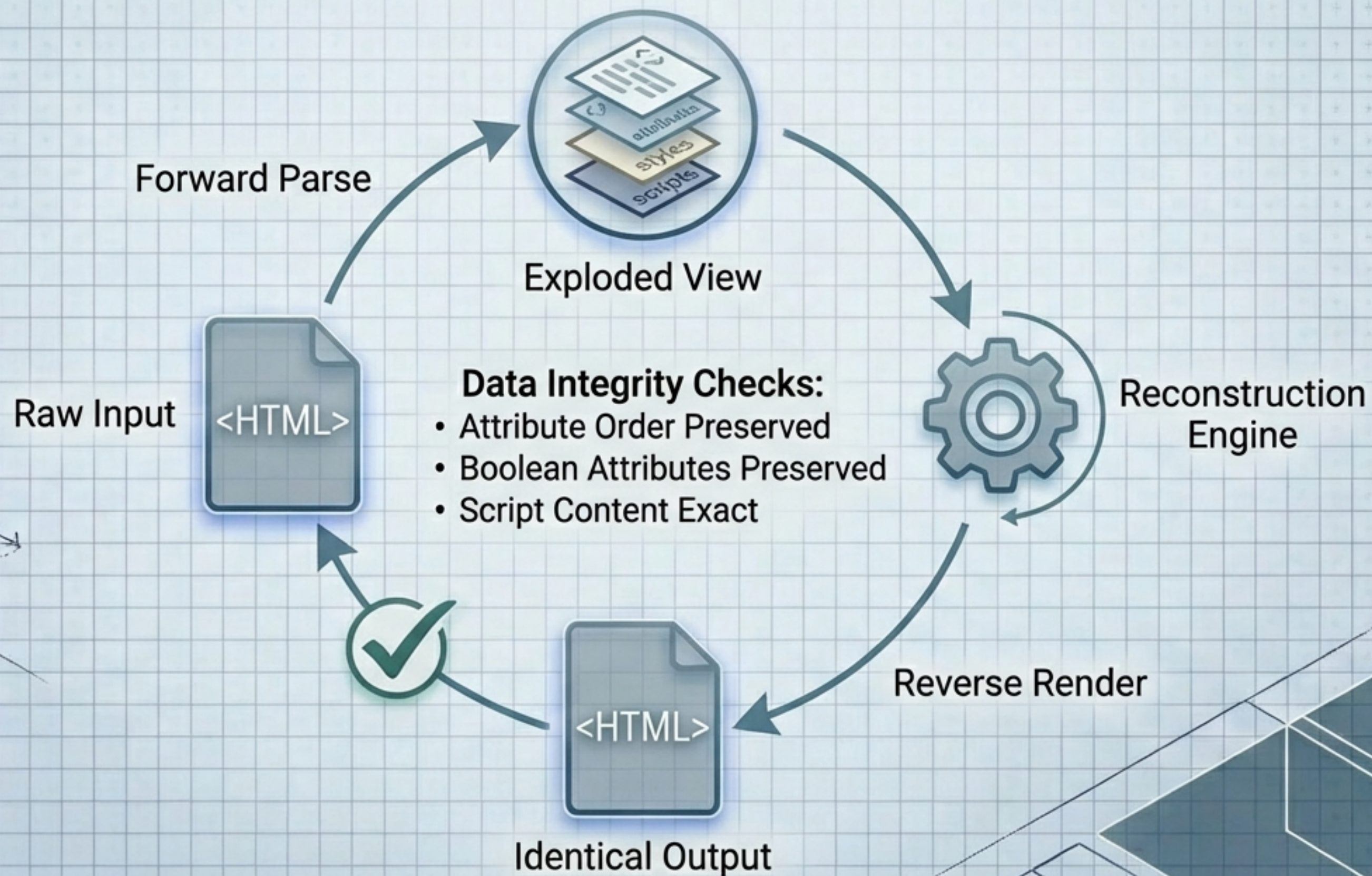
Styles Graph

```
...  
Node_Id: "node_56"  
Type: "text/css"  
Source: <link>  
Ref: "styles/main.css"  
...
```

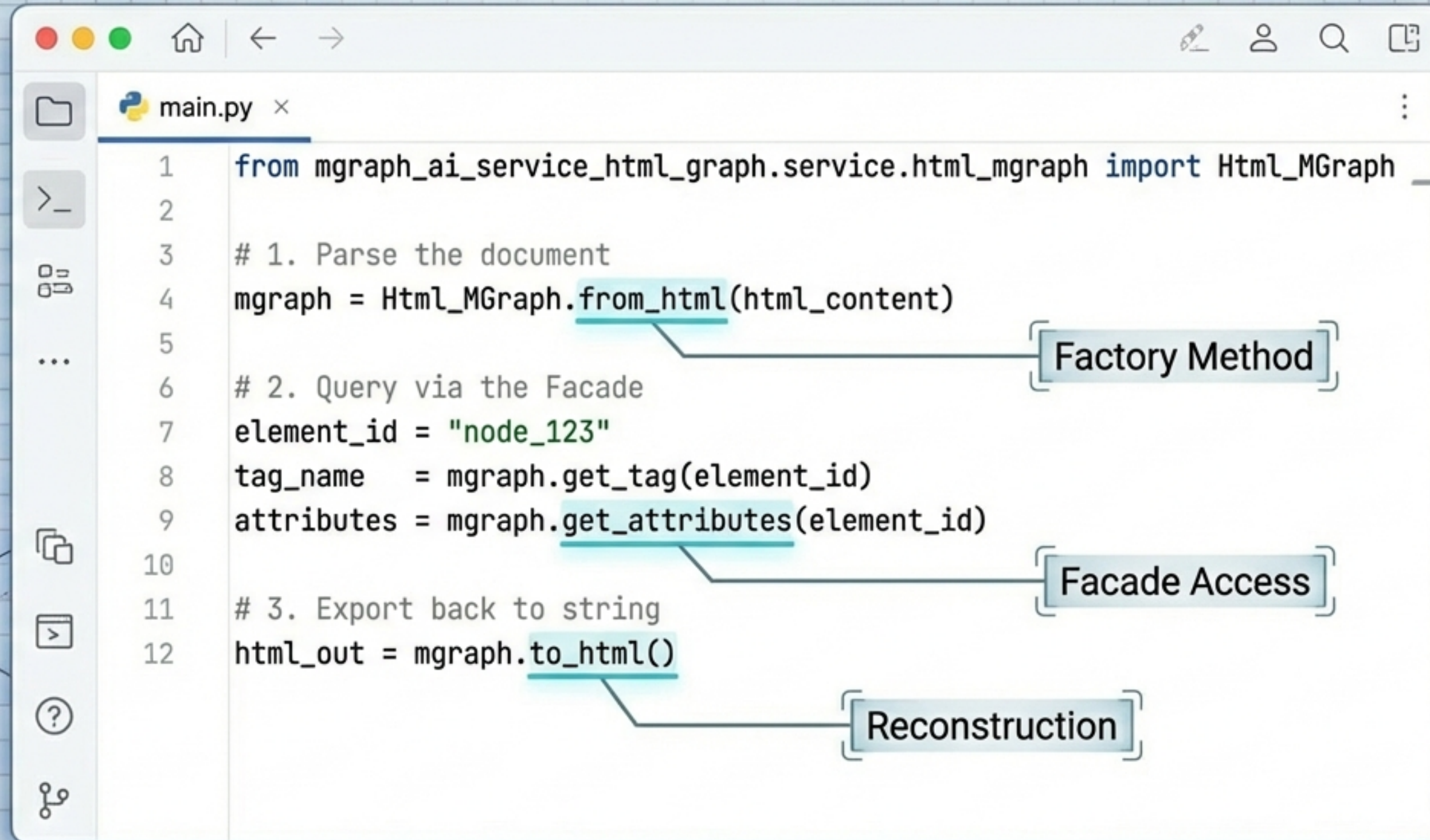
HTML

Analyze logic or extract design tokens without parsing the full DOM structure.

The Perfect Round-Trip Guarantee



Simplicity in Syntax



Surgical Precision: Common Patterns



Find elements by tag in `'body_graph'`, then check specific properties in `'attrs_graph'`.

Problem → Solution

Targeted Search



Extract all text content efficiently without traversing attribute nodes.

Problem → Solution

Content Extraction



Analyze `'<script>'` usage or build element info reports for security audits.

Problem → Solution

Audit & Analysis



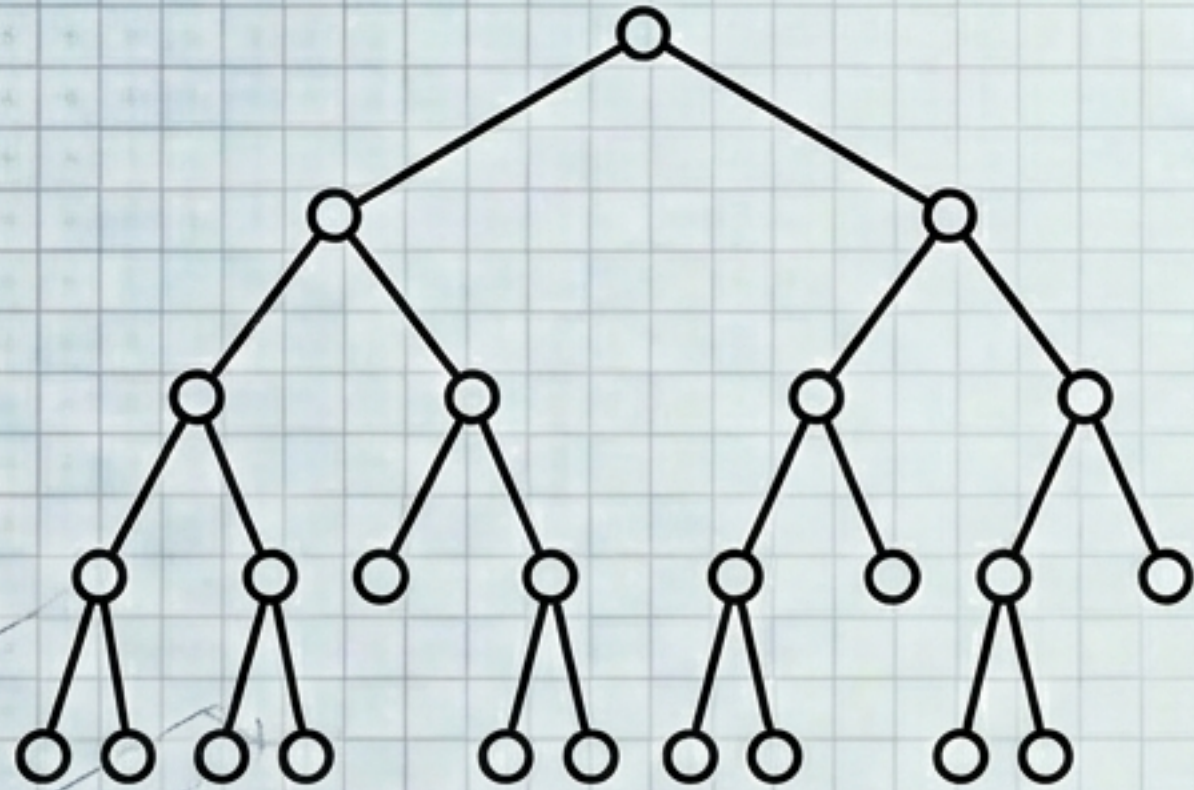
Execute round-trip transformation to verify data integrity.

Problem → Solution

Validation

Visualising the Data

Export to DOT (Graphviz)



```
mgraph.to_dot(graph='body')
```

Export to JSON / Dict

```
{  
  "node_id": "node_1",  
  "tag": "html",  
  "children": [  
    {  
      "node_id": "node_2",  
      "tag": "head"  
    },  
    {  
      "node_id": "node_3",  
      "tag": "body"  
    }  
  ]  
}
```

```
mgraph.to_json()
```


Developing with Html_MGraph

Best Practices (DO)



Use the Facade (`get_tag`) for simple operations.



Access component graphs (`body_graph`) for bulk operations.

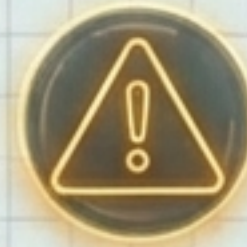


Check graph roots before iteration.

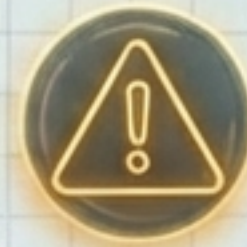


Utilize `mgraph.to_dot()` for debugging visualization.

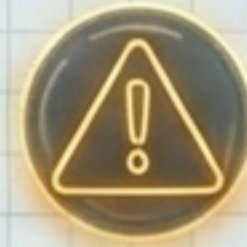
Anti-Patterns (DON'T)



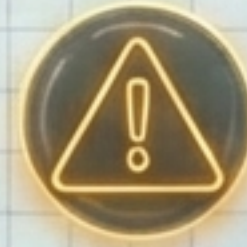
Assume an element exists in *all* graphs.



Modify graphs directly without using converters.



Rely on the string format of `Node_Id`.



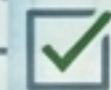
Ignore whitespace normalization in round-trips.

Resolving Common Friction

Issue
Detected

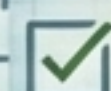
```
graph LR; A[Issue Detected] --- B[get_tag() returns None?]; A --- C[Round-trip HTML differs?]; A --- D[Attributes missing order?]; A --- E[Stats show zero?]; B --- F[Check: Is Node_Id registered in attrs_graph?]; C --- G[Check: Whitespace normalization or parser quirks.]; D --- H[Check: Use get_attributes() not dict iteration.]; E --- I[Check: Verify proper initialization.];
```

get_tag() returns None?



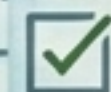
Check: Is Node_Id registered in attrs_graph?

Round-trip HTML differs?



Check: Whitespace normalization or parser quirks.

Attributes missing order?



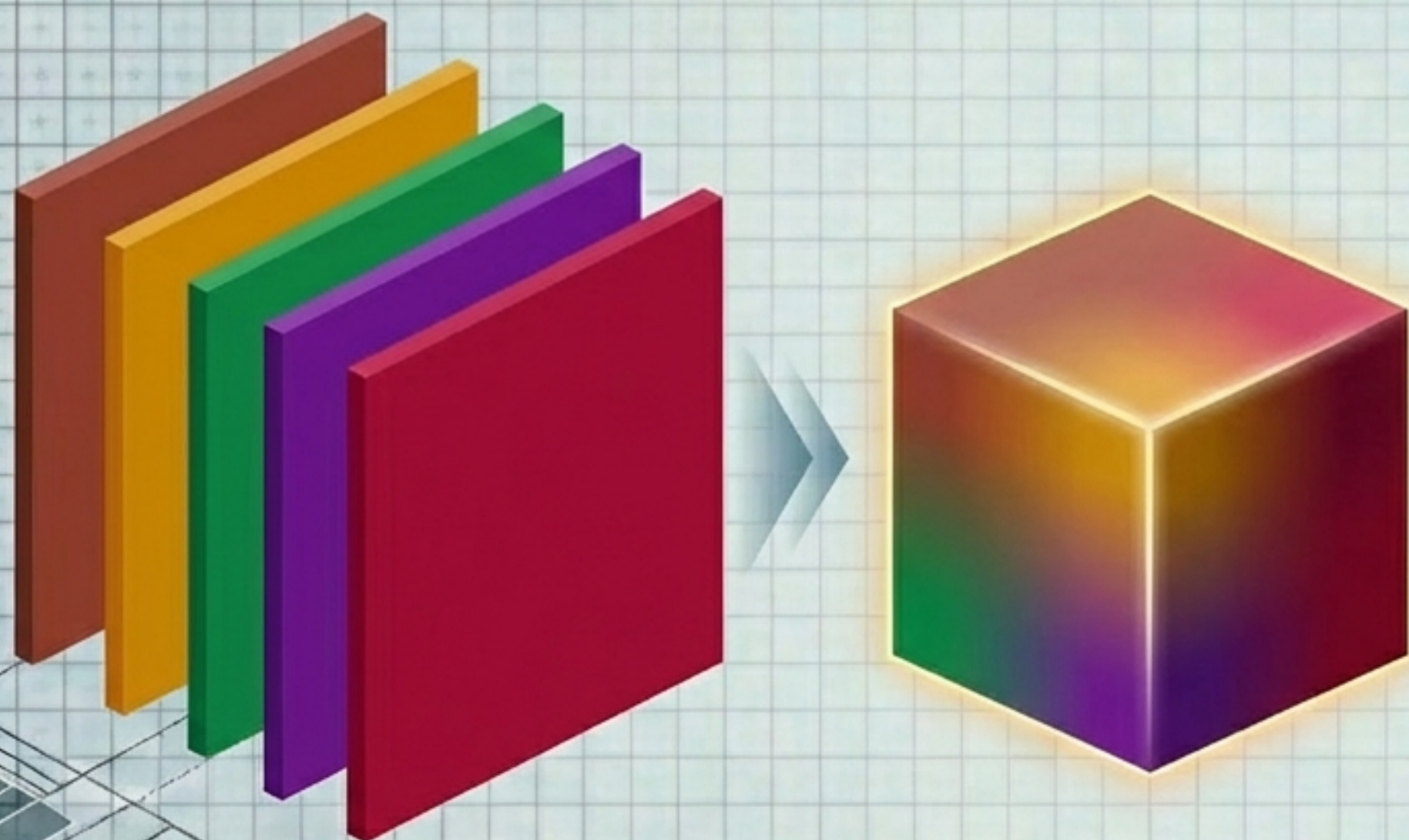
Check: Use get_attributes() not dict iteration.

Stats show zero?



Check: Verify proper initialization.

The MGraph Advantage



✓ **Normalised:** Complete separation of structure, data, and logic.

✓ **Type-Safe:** Secure foundation on MGraph-DB.

✓ **Interconnected:** Shared identity for seamless querying.

GitHub: `MGraph-AI_Service_Html_Graph`
Package: `pip install mgraph-db`